

BIG PINEY-MARBLETON, WY, USA

WMO: 726710

Lat: 42.584N

Lon: 110.108W

Elev: 2125

StdP: 78.27

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24164

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-23.8	-29.2	0.3	-26.4	-26.4	0.4	-23.2	11.8	-1.8	10.2	-3.1	1.4	80	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	30.1	13.1	28.6	12.8	27.2	12.6	15.1	25.3	14.2	24.6	13.5	24.1	5.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	11.4	17.3	10.6	10.3	16.7	9.1	9.3	16.3	49.3	25.0	46.5	24.4	44.1	23.8	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.3	8.3	-31.3	32.2	4.0	1.3	-34.1	33.1	-36.5	33.9	-38.7	34.6	-41.6	35.5
			-31.2	16.5	3.5	1.5	-33.7	17.5	-35.8	18.4	-37.8	19.2	-40.3	20.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-10.5	-8.8	-2.3	2.8	7.8	12.9	17.4	16.0	11.1	4.0	-3.6	-10.0
	DBStd	10.68	5.70	5.85	5.09	3.77	3.96	3.64	2.36	2.56	3.46	4.20	5.53	5.74
	HDD10.0	3112	636	527	383	216	90	16	0	1	28	187	407	621
	HDD18.3	5571	894	760	641	465	326	166	44	79	216	443	657	879
	CDD10.0	604	0	0	0	1	22	102	231	186	61	3	0	0
	CDD18.3	24	0	0	0	0	0	2	16	6	0	0	0	0
	CDH23.3	1607	0	0	0	0	15	233	774	494	89	1	0	0
	CDH26.7	399	0	0	0	0	1	49	233	107	8	0	0	0
Wind	WSAvg	3.1	2.4	2.5	3.1	3.8	3.7	3.8	3.3	3.1	3.0	3.0	2.6	2.3
Precipitation	PrecAvg	223	12	12	13	16	33	25	19	22	23	20	11	12
	PrecMax	359	25	25	32	32	96	104	46	74	45	66	22	34
	PrecMin	112	4	2	2	3	3	2	6	2	2	1	1	3
	PrecStd	69	7	7	8	9	22	24	10	14	12	14	5	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	8.2	15.8	20.2	25.2	30.4	32.4	31.2	27.8	22.1	14.1	7.2
		MCWB	0.5	2.2	5.2	7.5	11.1	12.6	14.4	14.0	12.5	9.6	5.1	1.4
	2%	DB	2.8	4.3	11.8	17.4	22.8	28.0	30.9	29.0	25.8	18.6	11.1	3.3
		MCWB	-0.8	0.4	3.6	6.3	9.9	12.3	13.5	12.4	11.0	7.7	3.8	-0.4
	5%	DB	0.6	2.3	8.7	14.8	20.9	26.2	29.2	27.8	24.0	16.2	8.0	1.2
		MCWB	-2.0	-0.6	2.5	5.1	9.1	11.6	13.1	12.6	10.2	6.4	2.5	-1.4
	10%	DB	-1.4	0.3	6.1	12.2	18.0	24.0	27.7	26.5	22.1	13.4	5.3	-1.1
		MCWB	-3.3	-2.0	1.3	4.2	7.9	10.9	13.0	12.0	9.7	5.3	0.9	-2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	2.5	5.9	8.4	12.7	14.3	16.8	16.1	14.2	10.1	6.1	1.9
		MCDB	4.6	6.8	13.0	19.0	23.5	27.1	28.3	26.9	23.9	19.4	12.8	6.0
	2%	WB	-0.4	1.0	4.3	6.8	10.6	13.2	15.5	14.8	12.8	8.6	4.2	0.2
		MCDB	2.2	4.0	10.6	15.8	20.5	25.1	25.9	24.6	22.0	16.5	9.8	3.0
	5%	WB	-1.9	-0.4	2.9	5.7	9.4	12.4	14.6	13.9	11.6	7.2	2.8	-1.2
		MCDB	0.4	1.9	8.1	13.6	18.8	23.7	25.1	23.7	20.4	14.4	7.5	0.9
	10%	WB	-3.4	-1.9	1.5	4.6	8.3	11.5	13.8	13.0	10.5	5.8	1.2	-2.9
		MCDB	-1.4	0.2	5.6	11.3	17.0	22.3	24.6	23.2	19.2	12.6	5.1	-1.0
Mean Daily Temperature Range	5% DB	MDBR	14.4	13.9	13.6	15.1	16.1	18.7	20.0	20.0	19.0	16.4	14.9	13.9
		MCDBR	15.2	13.8	17.3	20.5	21.2	22.9	22.6	22.6	23.0	21.4	17.7	14.7
		MCWBR	12.0	10.2	10.2	10.9	10.2	10.2	9.5	9.9	11.0	11.9	11.3	11.3
	5% WB	MCDBR	14.6	13.0	15.8	18.3	18.9	20.8	19.2	18.9	19.1	19.0	16.6	14.0
		MCWBR	11.8	9.9	9.8	10.2	9.4	9.7	8.6	8.8	9.4	10.8	10.9	10.9
Clear-Sky Solar Irradiance	taub	0.227	0.237	0.253	0.285	0.302	0.304	0.329	0.324	0.294	0.264	0.238	0.224	
	taud	2.387	2.372	2.442	2.444	2.475	2.505	2.453	2.456	2.535	2.590	2.553	2.452	
	Ebn at Noon	944	986	1005	986	970	965	936	933	946	946	931	921	
	Edh at Noon	76	92	99	107	107	105	109	105	90	74	65	65	
All-Sky Solar Radiation	RadAvg	2.11	3.00	4.34	5.48	6.19	7.39	7.26	6.28	5.08	3.42	2.25	1.71	
	RadStd	0.15	0.29	0.27	0.31	0.53	0.55	0.38	0.27	0.39	0.25	0.19	0.14	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page